

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

(192311208)

I G. Pallavi (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

G. Pallavi

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Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.
3. A smart traffic management system is being designed to control traffic signals across a city based on vehicle density, pedestrian movement, and weather conditions. The system must respond to real-time sensor inputs while also learning traffic patterns over time to reduce congestion. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for minimizing congestion and travel time.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Smart Home Automation System:-

A Smart Home Automation System uses Artificial Intelligence to automatically control lighting, temperature, and security based on user behaviour and environmental conditions. Intelligent agents collect information from sensors, make decisions, perform actions to improve comfort, energy efficient, and safety.

Types of Intelligent Agents:-

1. Simple Reflex Agent:-

A simple reflex agent makes decisions based on the current input using predefined rules.

Example:-

⇒ If motion is detected → Turn on the lights.

⇒ If temperature is above 30°C → Turn on the air conditioner.

⇒ If smoke is detected → Trigger the fire alarm.

2. Model-Based Reflex Agent:-

A model-based agent maintains an internal model of the environment. It remembers previous states and predicts what is happening even when information is incomplete.

Example:-

⇒ If motion was detected a few seconds ago, keeps hallway lights ON even if the sensor temporarily loses detection.

⇒ If a window is open while AC is running, it suggests closing the window to save energy.

3. Goal - Based Agent :-

A goal-based agent selects actions that help achieve specific goals.

Possible goals :-

- Maintain room temperature at 24°C .
- Reduce electricity consumption.
- Keep the house secure.
- Improve user comfort.

Example :-

when the user leaves home :

- Turn off unnecessary lights.
- Lock all doors.
- Activate Security Cameras.
- Switch appliances to power-saving mode.

4. Utility - Based Agent :-

A utility - Based Agent selects the action that provides the highest overall benefit by

Considering multiple factors.

Example :-

During Summer :

- Instead of running AC continuously,
- It balances temperature, electricity cost, occupancy, and weather conditions to choose the most efficient setting.

It considers :

- * Comfort.
- * Energy Consumption
- * Electricity cost
- * Security.

2. Robot in an Unknown Maze

A robot placed in an unknown Maze must find the exit without prior knowledge of the environment. Artificial Intelligence uses search algorithms and intelligent agents to help the robot explore efficiently.

1. Uniform cost Search (UCS):

Uniform cost Search expands the path with the lowest cumulative cost first.

Working :

1. Start at the initial position.
2. Calculate the path cost.

3. Expand the lowest-cost path.

4. Continue until the exit is found.

Example:-

⇒ If one path costs 8 units and another costs 5 units, UCS chooses the path costing 5 units first.

Complexity:-

- Time : $O(b^{\lceil (1+c^*/\epsilon) \rceil})$

- Space : $O(b^{\lceil (1+c^*/\epsilon) \rceil})$

2. Iterative Deepening Search (IDS):-

IDS combines the average of the advantages of Depth-first Search and Breadth-first Search.

Working:-

→ Search to depth 0.

→ If goal not found, Search to depth 1.

→ Continue increasing path depth until the exit is found.

Advantages

⇒ Very low memory usage.

⇒ Complete.

⇒ Suitable when solution depth is unknown.

Disadvantages

⇒ Repeats exploration

⇒ Can be slower due to repeated searches.

Complexity:-

- Time : $O(b^d)$

- Space : $O(b \times d)$